

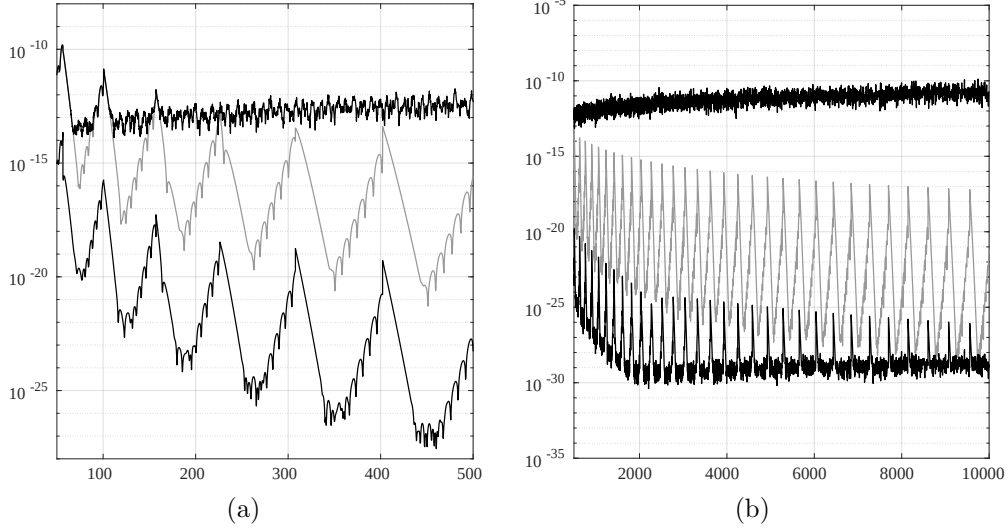


Figure 6: The values of $\Delta_p(t; 1/2, 2)$. The top curve corresponds to $p = 8$ implemented in double precision, the middle curve in gray color (the bottom curve) correspond to $p = 8$ and (respectively, $p = 12$) implemented in quadruple precision.

was chosen because $\zeta(s)$ does not grow too rapidly as $t \rightarrow +\infty$ in this strip, unlike in any strip where $\text{Re}(s) < 1/2$. The results of these computations are presented in Figure 6.

The top graph shows the error $\Delta_8(t; 1/2, 2)$, where $\zeta_8(s)$ was implemented in double precision (the benchmark values $\zeta(s)$ were computed in higher precision). We observe that rounding errors dominate the approximation error for $t > 200$. These rounding errors primarily arise when computing $\chi(s)$ and evaluating the values of n^{-s} and n^{s-1} for $s = \sigma + it$ with even moderately large t . Indeed, it is easy to verify numerically that when computing the value of 2^{it} in double precision for t of the order 10^2 , precision of the result will be around 10^{-14} , and when t increases to 10^3 , the precision drops to about 10^{-13} . This suggests that we lose one decimal digit of precision every time t increases by a factor of ten. Precision is also lost due to cancellation errors when we add many terms in the main sum in (3), though this effect likely plays a lesser role compared to rounding errors.

The middle graph in Figure 6 (shown in gray) displays the errors $\Delta_8(t; 1/2, 2)$, where this time $\zeta_8(s)$ was implemented in quadruple precision. We observe that rounding errors do not affect the results in this range of t . The quadruple precision implementation of $\zeta_8(s)$ produces errors smaller than 10^{-13} for $t > 250$ and smaller than 10^{-15} for $t > 2000$ in the strip $1/2 \leq \sigma \leq 2$. Of course, when t becomes very large (of the order 10^{10} or greater), rounding errors will eventually become noticeable.

The bottom graph on Figure 6 shows the errors $\Delta_{12}(t; 1/2, 2)$ for the quadruple precision implementation of $\zeta_{12}(s)$. These errors are significantly smaller: we find that $\Delta_{12}(t; 1/2, 2) < 10^{-25}$ for $t > 5000$. The effects of rounding errors become clearly noticeable for $t > 2000$, though they do not pose a significant issue in this range, as the maximum approximation error remains larger than the rounding errors. However, when t reaches 10^5 or greater, rounding errors will dominate the approximation error, and the graph will resemble the top graph (though with a much smaller overall error, of order 10^{-27}).

Now that we have (hopefully) convinced the reader of the accuracy of our approximations, it is time to discuss how they were derived and to present an algorithm for computing the coefficients $\omega_{p,j}$ and $\lambda_{p,j}$.